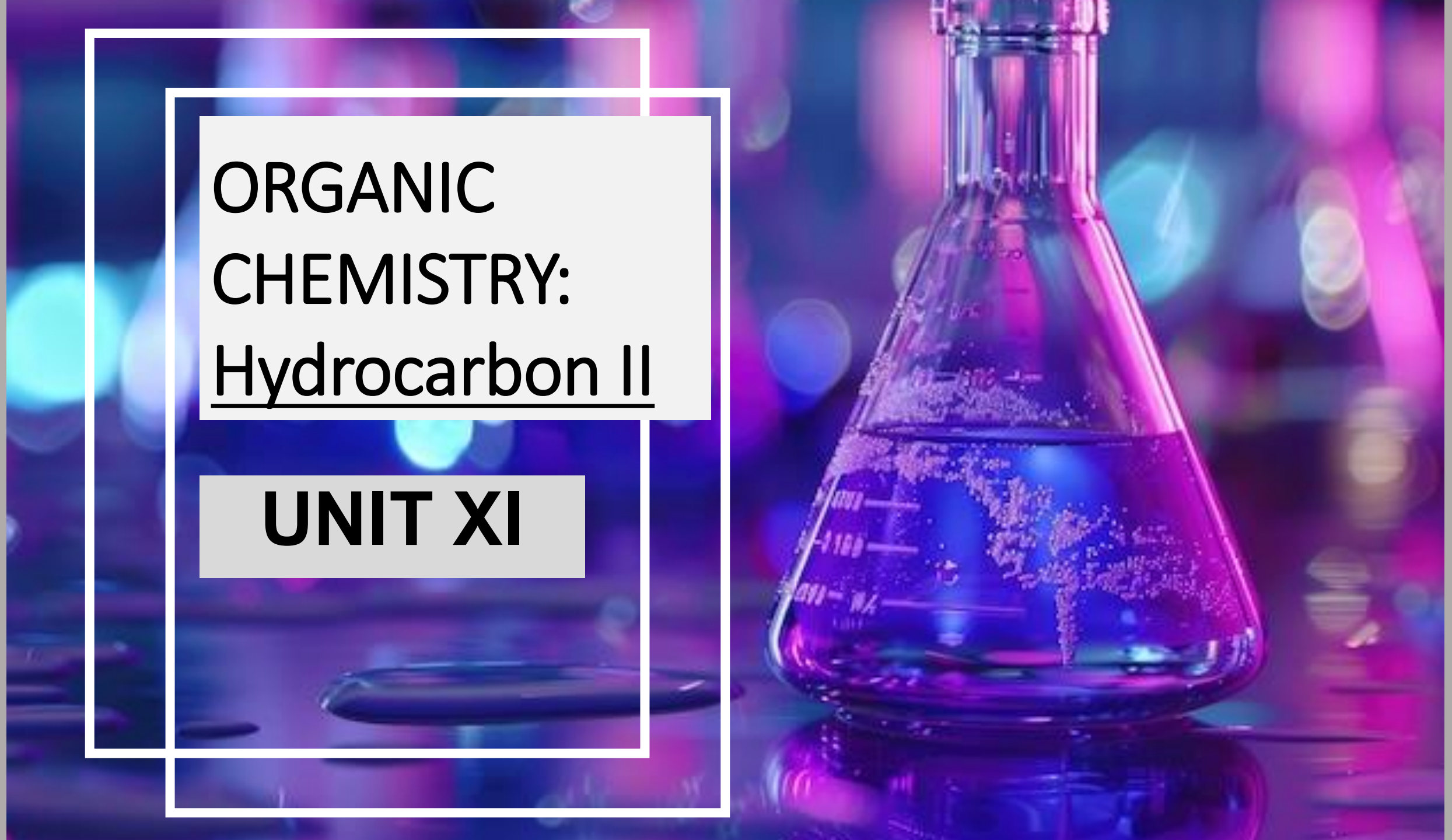


ORGANIC CHEMISTRY: Hydrocarbon II

UNIT XI

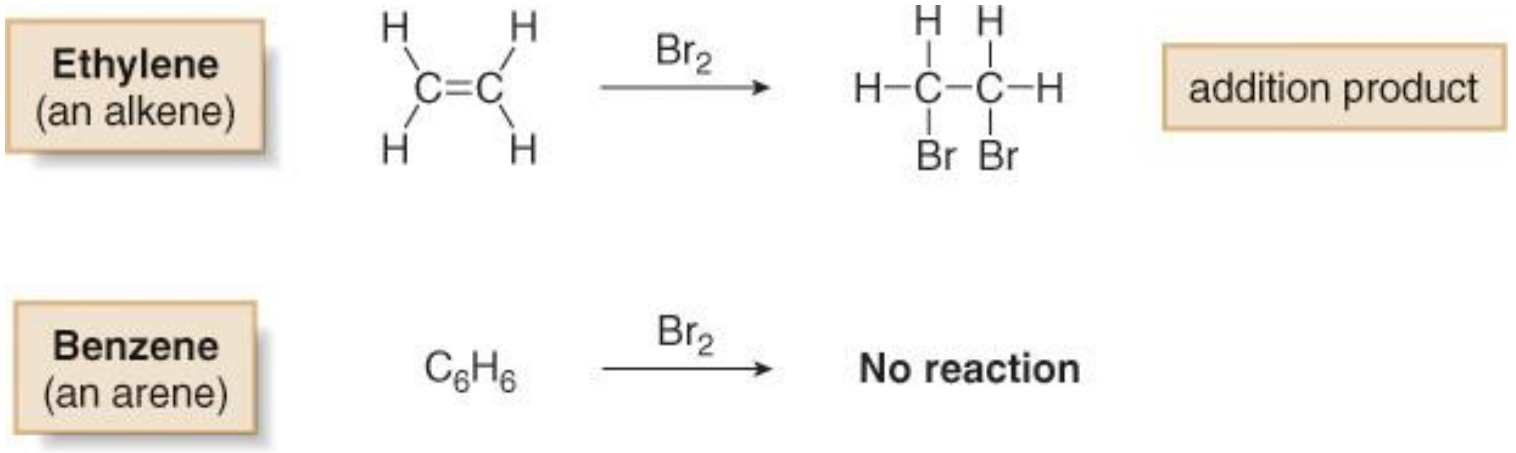
Learning Outcomes

At the end of each topic, the students should be able to:

- Identify and group hydrocarbons (alkanes, alkenes, alkynes, cycloalkanes, and aromatics) by their structure and bonds.
- Name and write the structural formulas of hydrocarbon using IUPAC nomenclature rules.
- Explain the physical properties (such as boiling points, melting points, and solubility) of hydrocarbons.
- Describe the characteristic chemical reactions of hydrocarbons.

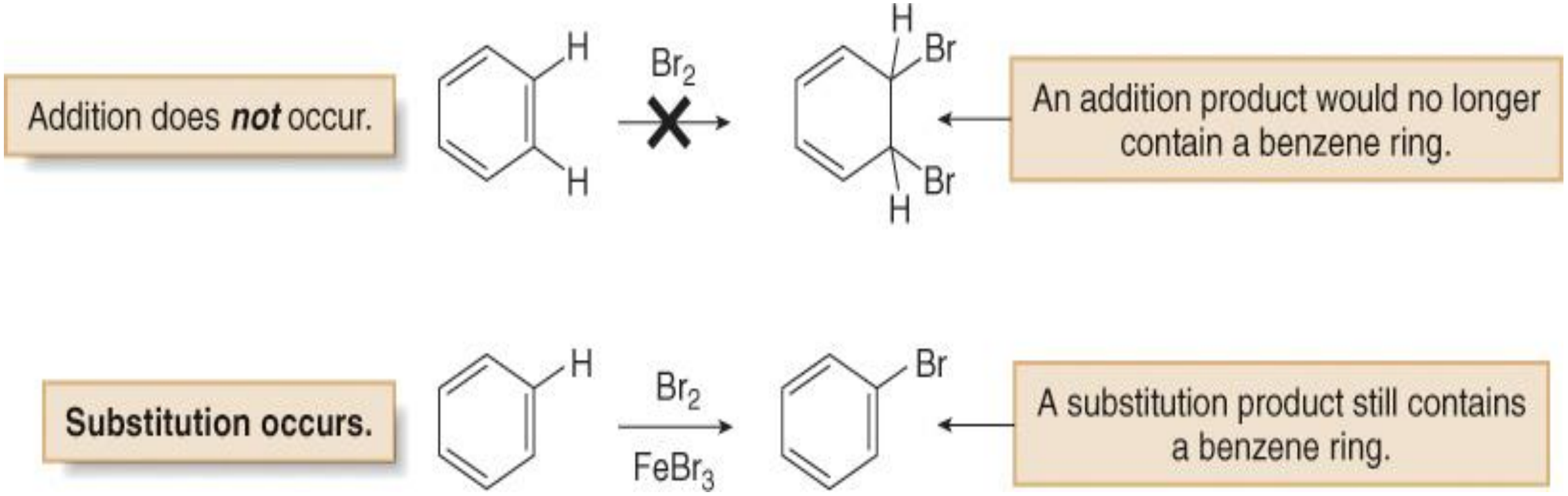
Benzene and Aromatic Compounds

- Benzene (C_6H_6) is the simplest aromatic hydrocarbon (or arene).
- First isolated by Michael Faraday but structure was proposed by August Kekule after almost 50 yrs.
- Benzene has three degrees of unsaturation, making it a highly unsaturated hydrocarbon.
- Whereas unsaturated hydrocarbons such as alkenes, alkynes and dienes readily undergo addition reactions, benzene does not.



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- Benzene does react with bromine, but only in the presence of $FeBr_3$ (a Lewis acid), and the reaction is a substitution, not an addition.



- At the same time, Benzene does not also react with Bayer's Test giving a diol product.

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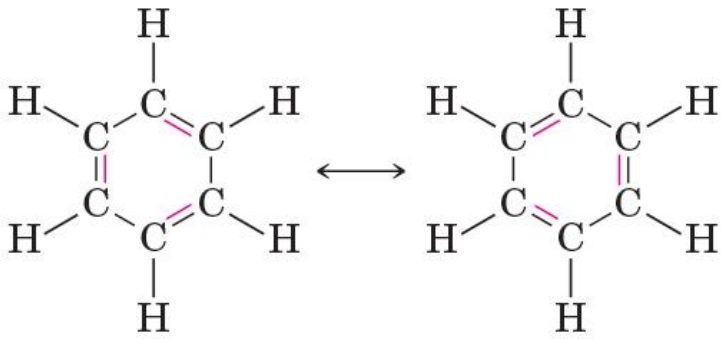
Resonance structure for benzene (1930s)

- The theory of resonance developed by Linus Pauling provided the first adequate description of the structure of benzene.
- According to the theory of resonance, certain molecules and ions are best described by writing two or more Lewis structures. The real molecule or ion is a **resonance hybrid** of these structures.
- Each individual Lewis structure is called a **contributing structure.**
- We show that the real molecule is a resonance hybrid of the two or more Lewis structures by using a double-headed arrow between them.

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Benzene

- Here are two contributing structures for benzene:



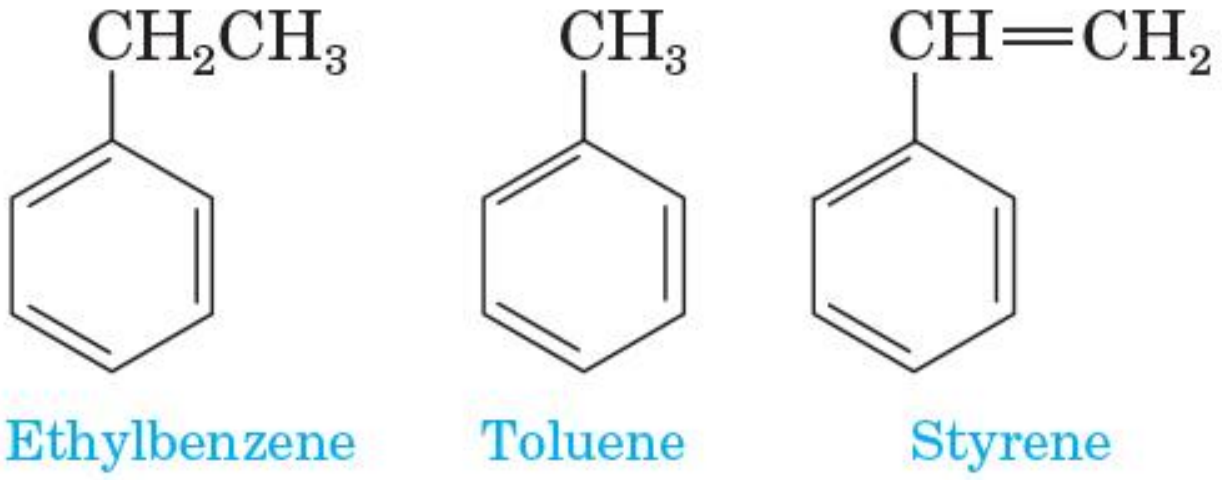
Alternative Lewis contributing structures for benzene

- The resonance hybrid has some of the characteristics of each Lewis contributing structure.
- The length of a carbon-carbon bond in benzene, for example, is midway between that of a carbon–carbon single bond and a double bond.

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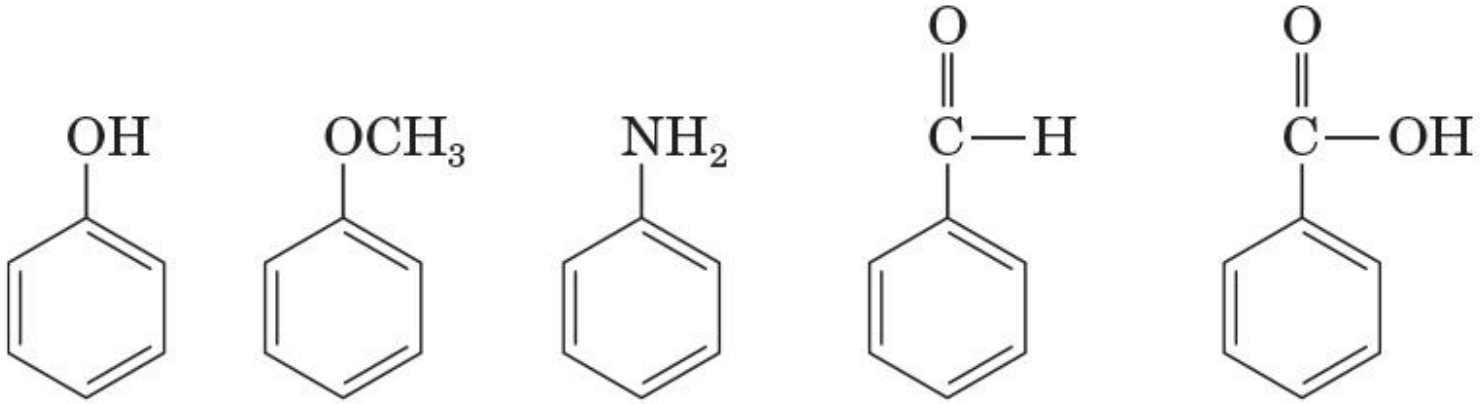
Nomenclature

- Monosubstituted alkylbenzenes are named as derivatives of benzene; for example, ethylbenzene.
- The IUPAC system retains certain common names for several of the simpler monosubstituted alkylbenzenes.



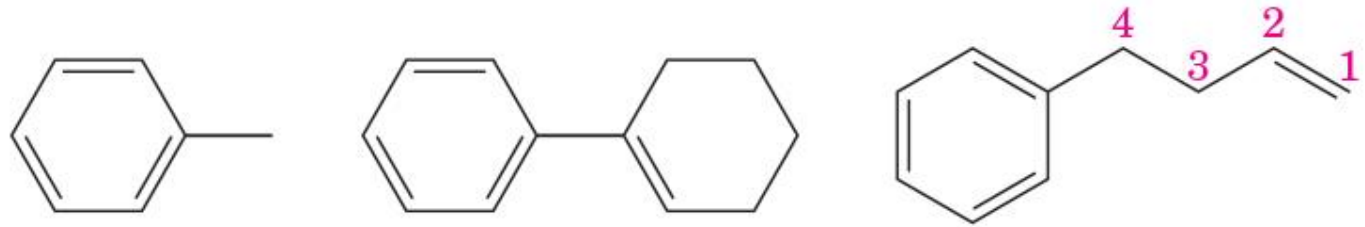
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- The common names for these monosubstituted benzenes are also retained



Phenol | Anisole | Aniline | Benzaldehyde | Benzoic acid

- **Phenyl group (C_6H_5– or Ph–):** The substituent group derived by removal of an H from benzene.

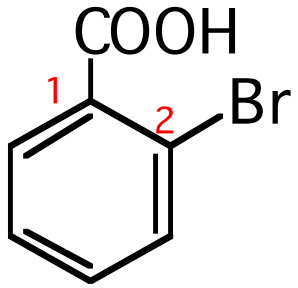


Phenyl group (C_6H_5—; Ph—) | 1-Phenylcyclohexene | 4-Phenyl-1-butene

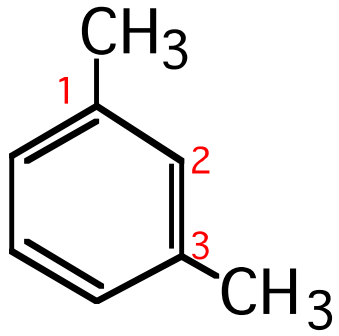
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When two substituents occur on a benzene ring, three isomers are possible; they may be located by:

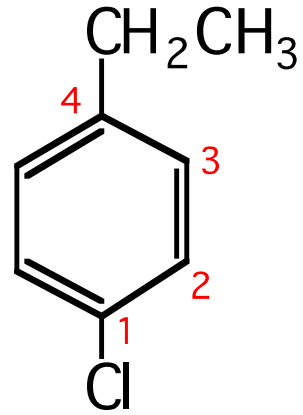
- numbering the atoms of the ring or
- using the locators ortho (*o*), meta (*m*), and para (*p*).



2-Bromobenzoic acid
(*o*-Bromobenzoic acid)



1,3-Dimethylbenzene
(*m*-Xylene)

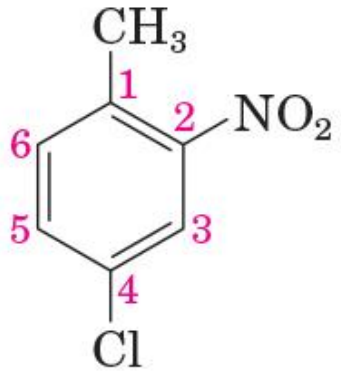


1-Chloro-4-ethylbenzene
(*p*-Chloroethylbenzene)

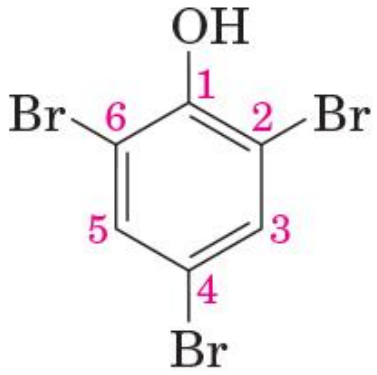
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For three or more substituents

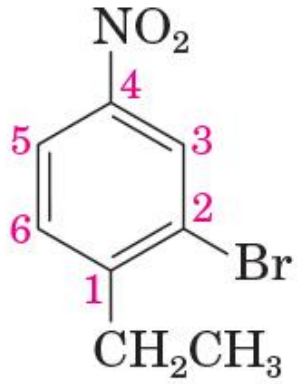
- If one of the substituents imparts a special name, name the molecule as a derivative of that parent.
- If none of the substituents imparts a special name, number the substituents to give the smallest set of numbers, and list them in alphabetical order before the ending "benzene."



4-Chloro-2-nitrotoluene



2,4,6-Tribromophenol

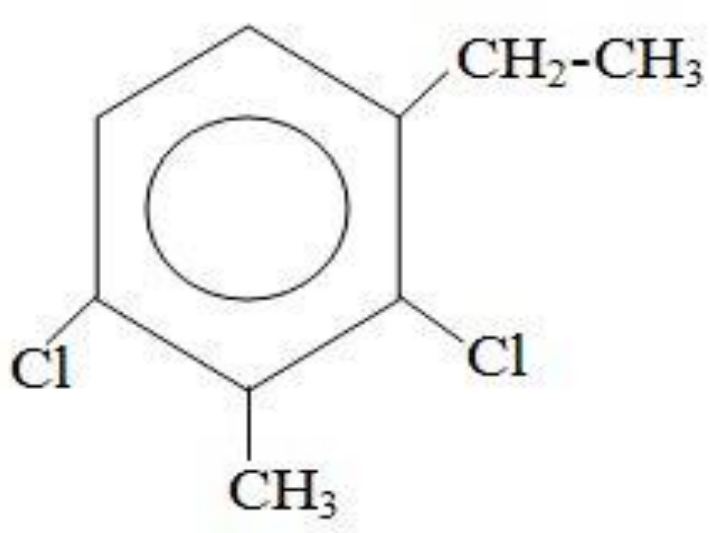


2-Bromo-1-ethyl-4-nitrobenzene

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EXERCISE

What is the correct IUPAC name of the following aromatic compound?



a. 1,3-dichloro-4-ethyl-2-methylbenzene

b. 1,5-dichloro-4-ethyl-6-methylbenzene

c. 2,6-dichloro-3-ethyltoluene

d. 2,6-dichloro-5-ethyltoluene

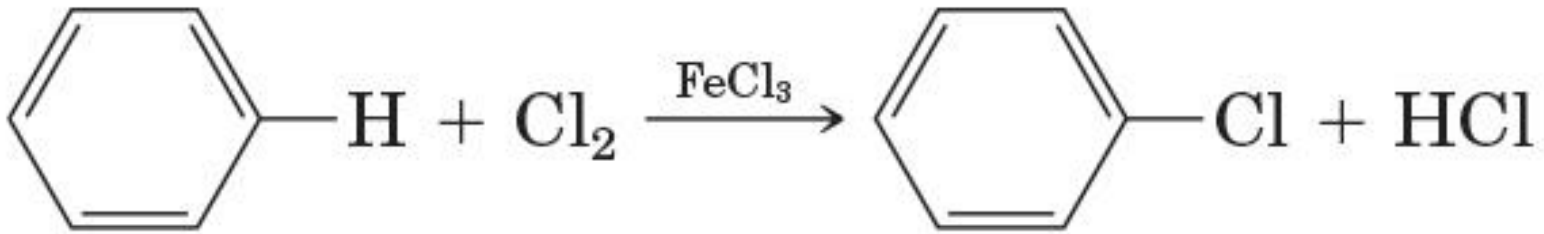
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Reactions of Benzene

By far the most characteristic reaction of aromatic compounds is substitution at a ring carbon.

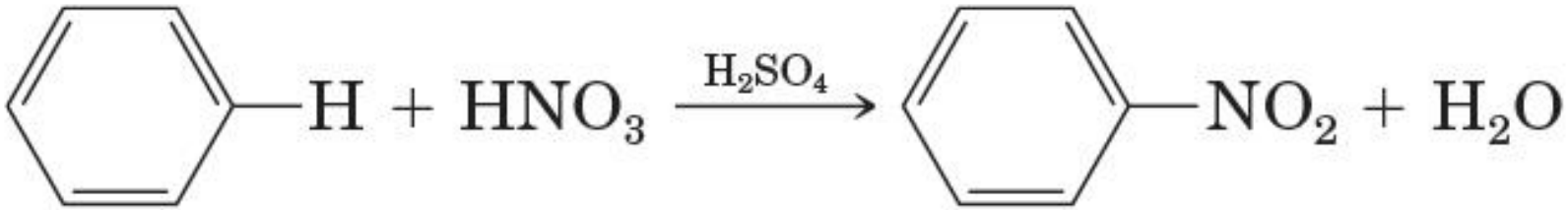
- This reaction is called **aromatic substitution**.
- Some groups that can be introduced directly on the ring are the halogens, the nitro (–NO_2) group, and the sulfonic acid (–SO_3H) group.

Halogenation:



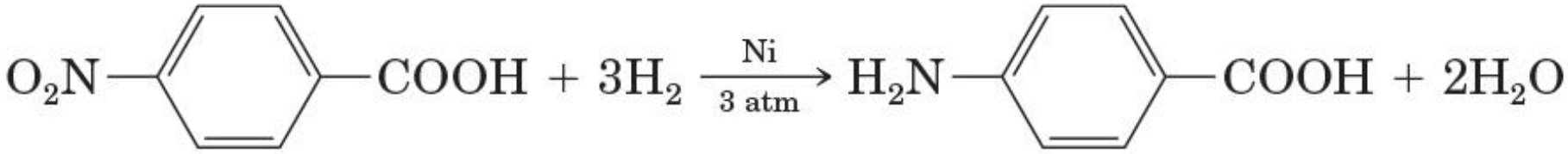
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Nitration:



Nitrobenzene

- A value of nitroarenes is that the nitro group can be reduced to a primary amino group.

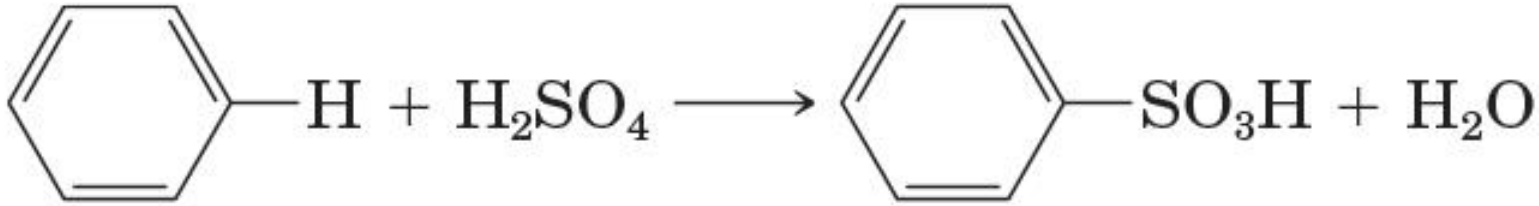


4-Nitrobenzoic acid
(*p*-Nitrobenzoic acid)

4-Aminobenzoic acid
(*p*-Aminobenzoic acid, PABA)

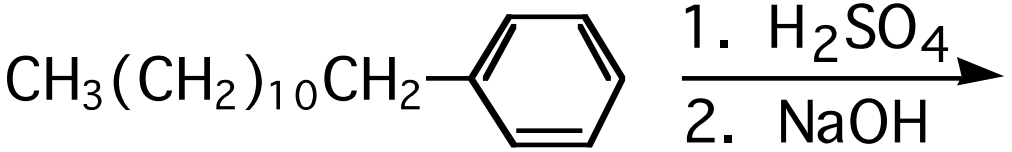
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Sulfonation:

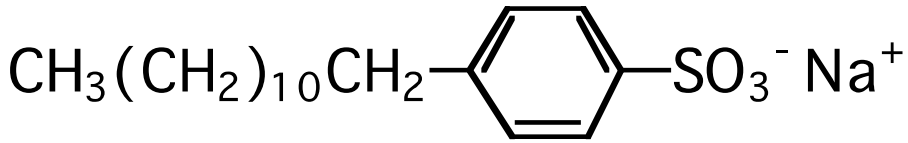


Benzenesulfonic acid

- An application of sulfonation is in the preparation of synthetic detergents.



Dodecylbenzene

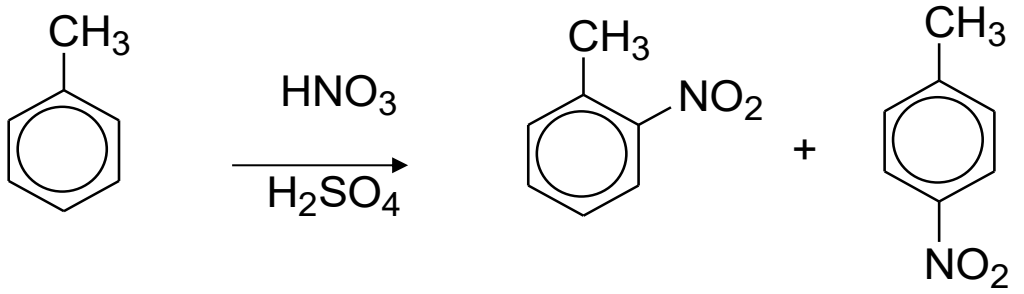


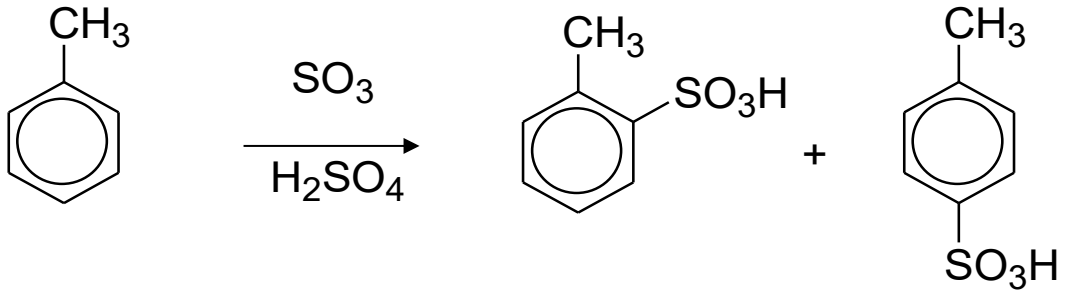
Sodium 4-dodecylbenzenesulfonate, SDS (an anionic detergent)

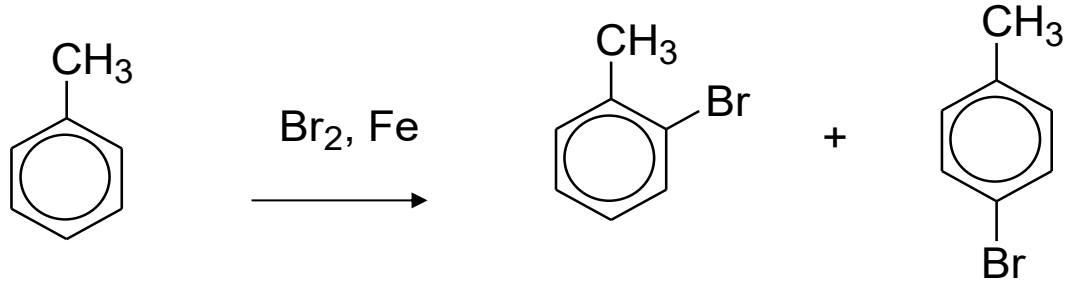
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Reactions of Monosubstituted Aromatic Compounds

Toluene



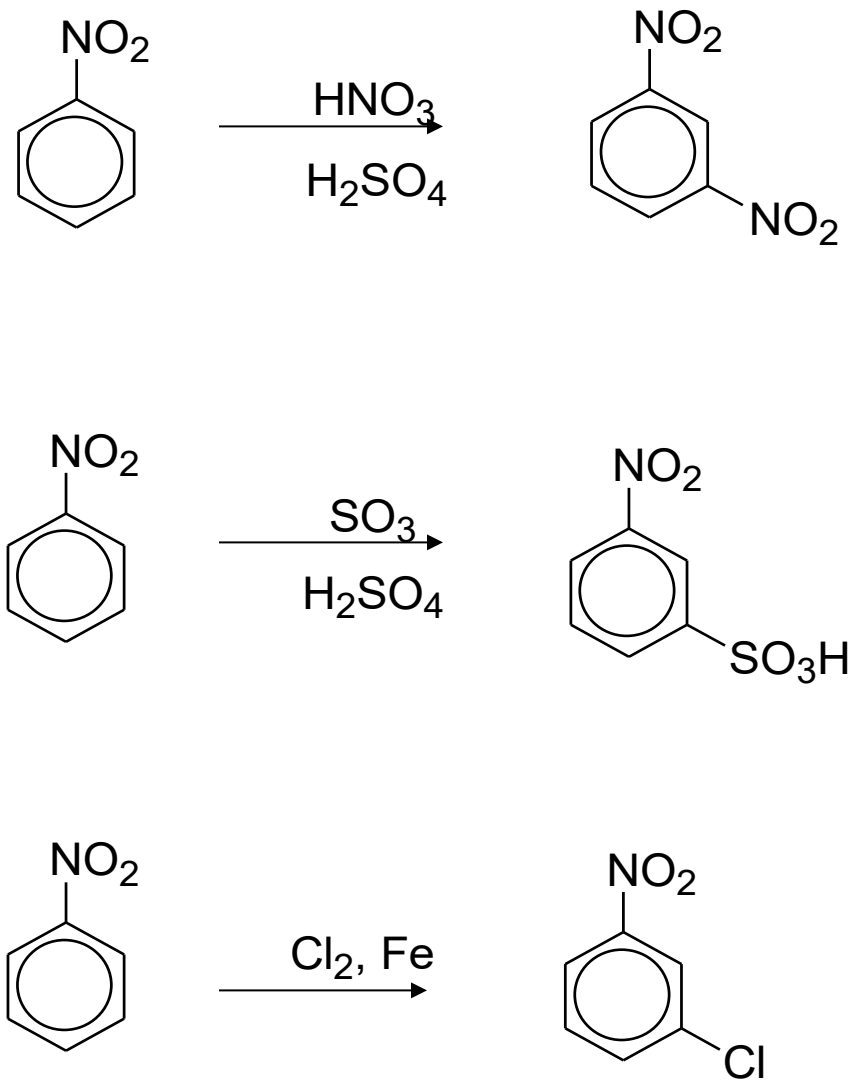




- faster than the same reactions with benzene

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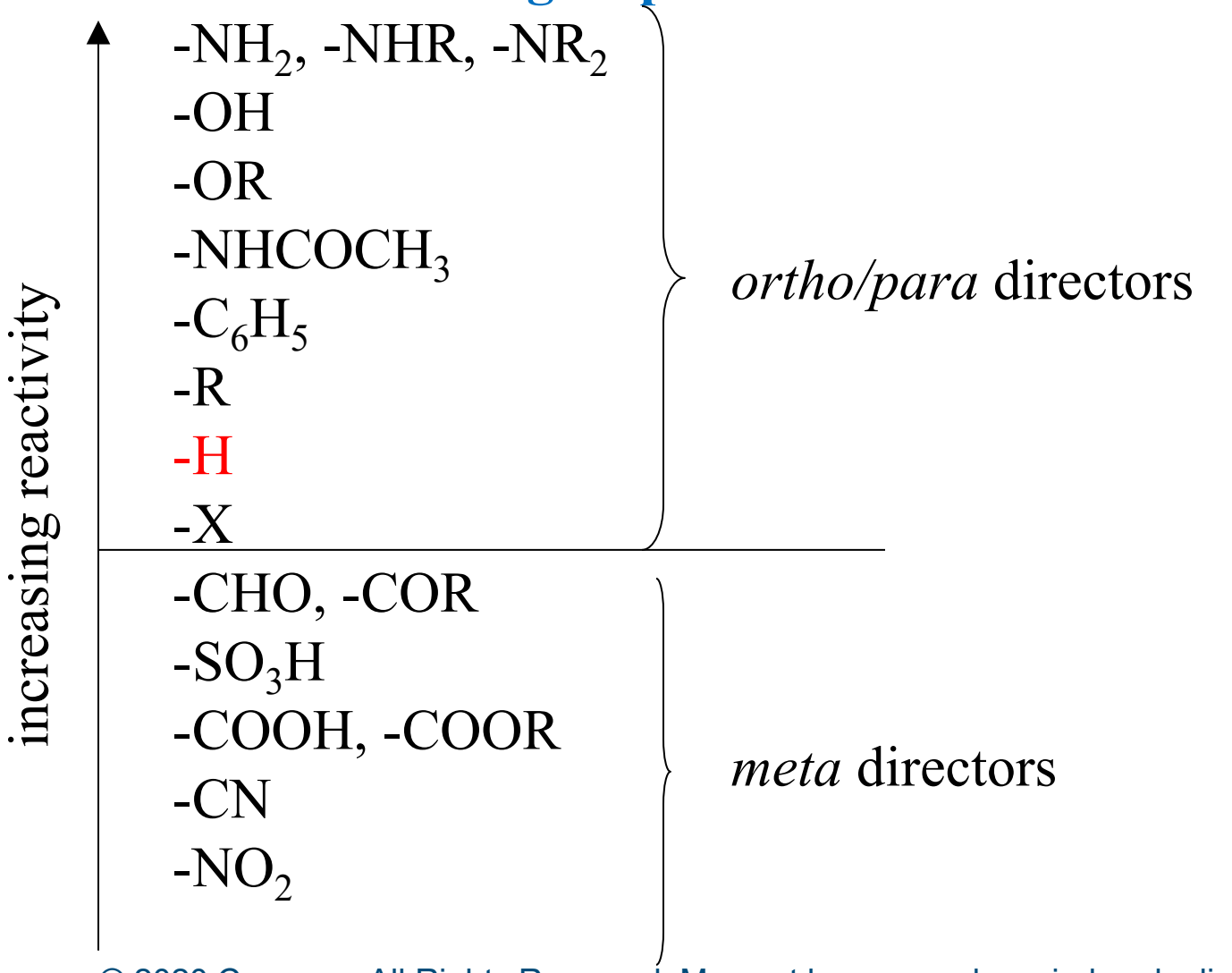
Nitrobenzene



- slower than the same reactions with benzene

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Common substituent groups and their effect on EAS:



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EXERCISE

Aromatic hydrocarbons undergo two types of _____ reactions known as _____ and _____.

a. Addition; alkylation; hydrohalogenation

b. Addition; alkylation; halogenation

c. Substitution; alkylation; hydrohalogenation

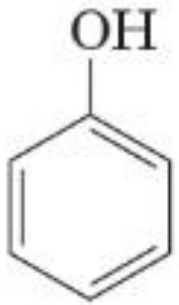
d. Substitution; alkylation; halogenation

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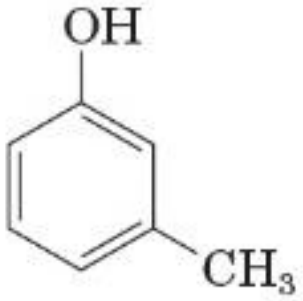
Phenols

The functional group of a phenol is a hydroxyl (–OH) group bonded to a benzene ring.

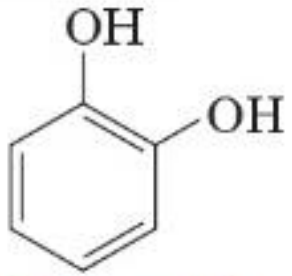
- ❑ Name substituted phenols either as derivatives of phenol or by common names.



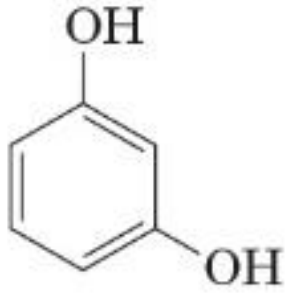
Phenol



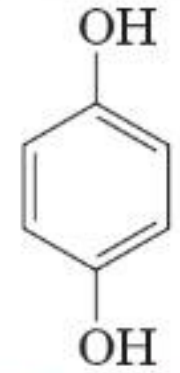
3-Methylphenol
(*m*-Cresol)



1,2-Benzenediol
(Catechol)



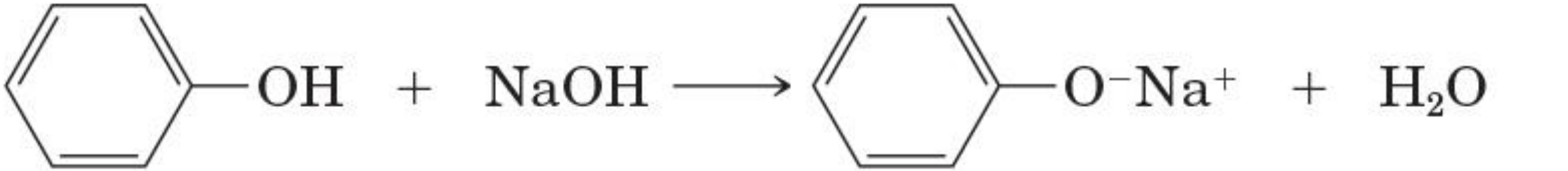
1,3-Benzenediol
(Resorcinol)



1,4-Benzenediol
(Hydroquinone)

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- Most phenols are weak acids, with pK_a values of approximately 10.
- They are insoluble in water but react with strong bases, such as NaOH and KOH to form water-soluble salts.



Phenol
$pK_a = 9.89$
(stronger acid)

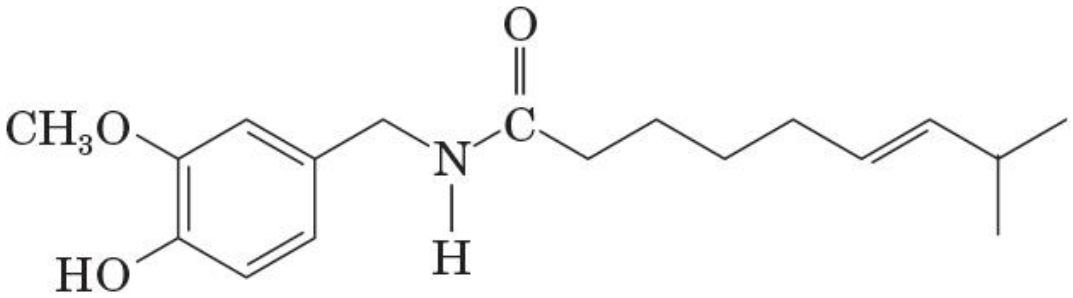
Sodium
hydroxide
(stronger base)

Sodium phenoxide
(weaker base)

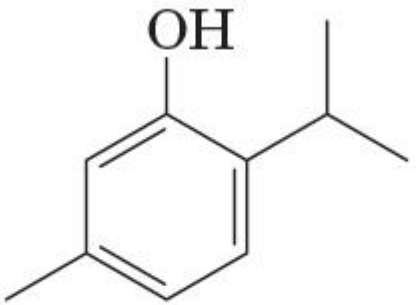
Water
$pK_a = 15.7$
(weaker acid)

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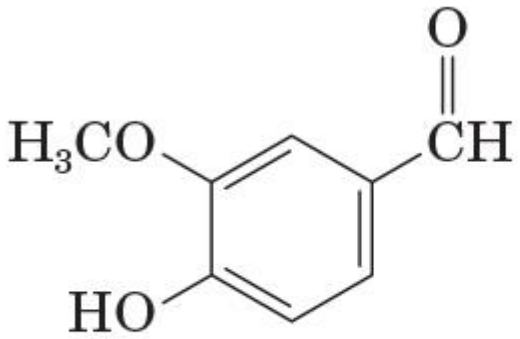
Some phenols found in nature.



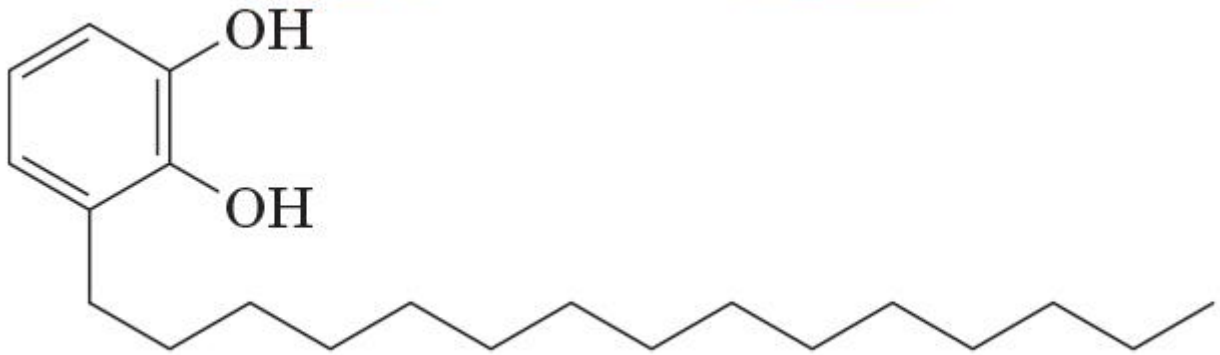
Capsaicin
(from various types of peppers)



2-Isopropyl-5-methylphenol
(Thymol)



4-Hydroxy-3-methoxy-benzaldehyde
(Vanillin)



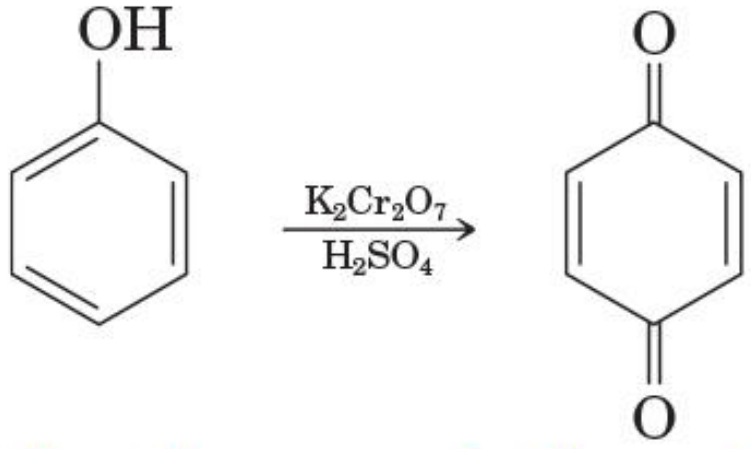
Urushiol

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Oxidation of Phenols

Phenols are oxidized to quinones by a variety of strong oxidizing agents.

- Phenol is oxidized by potassium dichromate in the presence of sulfuric acid to form *p*-quinone.



Phenol

1,4-Benzenedione
(*p*-Quinone)

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EXERCISE

What is the name of the simplest organic compound with an —OH attached to a carbon atom in the aromatic ring?

a. Phenyl alcohol

b. Benzyl alcohol

c. Cyclohexyl alcohol

d. Phenol

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EXERCISE

One of the most important properties of phenols is their:

a. acidity.

b. basicity.

c. reactivity.

d. combustion.

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EXERCISE

Identify the uses of phenol or phenol derivatives.

a. Antiseptic in mouthwash and throat lozenges

b. Active ingredient in Lysol

c. Antioxidant food additive

d. All the above

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